01 **BRIEFING NOTE**

ESCAP

APCICT
Asian and Pacific Training Centre for Information and Communication Technology for Development

The Ethics of AI

KEY TOPICS

Trustworthy AI:
Ethical groundwork for AI

Information Society and AI:
Issues of AI in the networked society that complicates ethical concerns

Intelligent Information Society and Ethics:
Technical and social aspects of AI that should be considered in the ethical framework

Dual Role of Government
Responsibilities for promoting the creation and broad dissemination of technologies that serve the needs and interests of humanity.

Overview

Recent notable milestones like AlphaGo and the emergence of ChatGPT highlight the significance of AI (artificial intelligence). However, AI models are growing exponentially, the number of parameters reaching trillions, whereas humans typically utilize only 10% of 100 trillion synapses. As AI advances, it presents ethical challenges to effectively harness its advantages.

In developing countries, some main barriers are limited resources and investments, insufficient funding, and a lack of knowledge[1] and expertise[2] in AI and ML (machine learning). These can result in a brain drain to developed countries, as workers may leave for better opportunities in other countries investing in AI. To overcome the challenges, it is crucial to ensure efficient and ethical development and implementation of AI from the beginning, and understand its potential threats, limitations, and societal requirements.

To integrate AI efficiently for the betterment of humanity and ensure trust as a foundation, it is essential to focus on making AI systems and algorithms public, accountable, controllable, and transparent, along with increasing public education and awareness. Additionally, it is important that AI solutions are aligned with local values to effectively address localized issues.

[1] A. Ojo, S. Mellouli, and F. Ahmadi Zeleti, "A realist perspective on AI-era public management," in Paper presented at the 20th annual international conference on digital government research, Dubai United Arab Emirates, 2019.
[2] O. S. Al-Mushayt, "Automating E-government services with artificial intelligence," IEEE, 2019.

Establishing an ethical framework for AI involves

- Highlighting the importance of trustworthy AI,
- Addressing the issues of AI in the information society that complicates ethical concerns,
- Incorporating technical and socio-economic concerns into the ethical framework in the intelligent information society,
- Providing insights into unique contexts for developing countries.

Trustworthy AI

The Importance of Ethics and Traditional Ethics

AI plays a crucial role in enhancing efficiency, analyzing large amounts of data beyond human capabilities, promoting innovation, streamlining processes, and improving customer experience. It facilitates the achievement of SDGs by promoting economic growth, reducing poverty, enhancing health and education outcomes, and mitigating climate change.

Since AI has profound and pervasive impacts across all aspects of society and its progress is irreversible, human trustworthiness is essential as AI is created and shaped by humans. While AI lacks human-like comprehension and emotional intelligence, trust in AI systems is established through assessing performance, such as reliability, transparency, accuracy, consistency, and predictability. As traditional ethics are grounded in human morality and accountability, traditional ethics are essential in creating unbiased AI algorithms, protecting user privacy, utilizing AI for societal progress, and ensuring transparent and accountable decision-making. However, traditional ethics encounters challenges due to the following factors: being constrained by time and place, prioritizing the motivations behind actions rather than their outcomes, and having a human-centric focus that overlooks non-human entities.

In applying traditional ethics on AI ethics, local culture should be considered, since traditional ethics can vary across cultures and thus provide different guidelines due to context.

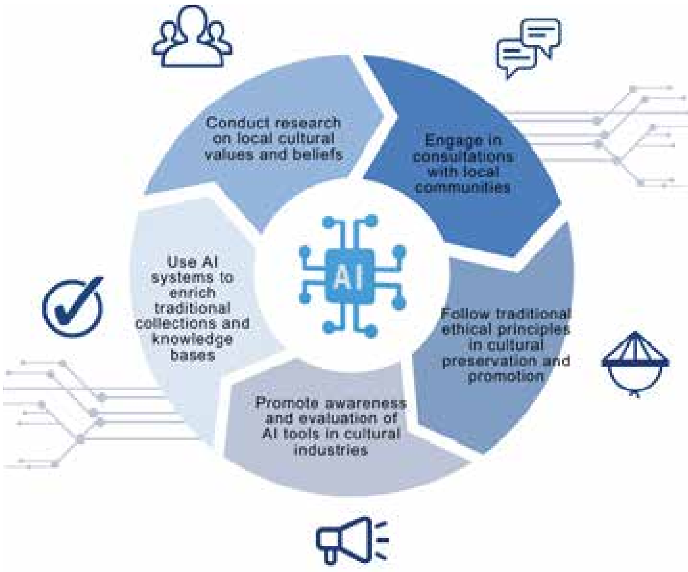

Fig. 1 Embedding Cultural Values in AI tools

To integrate those values, below are some recommendations:

- Conduct research on local cultural values and beliefs,
- Engage in consultations with local communities,
- Adhere to traditional ethical principles in cultural preservation and promotion,
- Promote awareness and evaluation of AI tools in cultural industries,
- Use AI systems to enrich traditional collections and knowledge bases.

AI Ethics for All Stakeholders

Users play a crucial role in ensuring ethical AI practices. The incident of Microsoft's chatbot Tay highlighted the importance of user input. Moreover, any consumer can easily be a developer due to open-source platforms. Therefore, user judgement, awareness, feedback and involvement are essential for ethical and beneficial AI systems.

Key stakeholders for AI are as follows:

- Consumers: End-users concerned about privacy, fairness, and accuracy.
- Workers: Employees impacted by AI deployment in the workplace.
- Developers: Responsible for reliable and ethical AI development.
- Investors: Concerned about financial and reputational risks of unethical AI practices.
- Civil society organizations: Advocate for ethical development and use of AI.
- Regulators: Government bodies overseeing AI development and enforcing ethical policies.

ESG (environmental, social, and governance) Management and AI Ethics

ESG examines a company's impact on the environment and stakeholders, and ensures ethical standards and governance practices are in place. Likewise, AI ethics concerns the responsible and ethical concerns from the development and usage of AI. By integrating ESG management into the development of AI, businesses and organizations can ensure that their AI systems are not only technologically advanced, but also ethically responsible and in line with societal values. Energy consumption serves as an example of how ESG is integrated into AI development, with developers employing efficient algorithms to minimize energy usage.

Recommendations to harness AI for Environmental Solutions are:

- Creating incentives for ethical and rights-based AI solutions.
- Fostering collaboration between environmental experts.
- Involving local and indigenous communities in AI development.

Lack of Regulation and Ethics

Assigning responsibility within the legal system is one of the biggest challenges in AI.

The lack of regulation in AI development is attributed to:

- Inadequate capacity and expertise in creating regulatory frameworks,
- Limited awareness among regulators regarding AI risks and consequences,
- Resistance from the AI industry fearing hindered innovation and growth,
- Challenges in adapting regulations from other countries due to cultural, social, and political differences,
- Insufficient international cooperation.

The consequences of the lack of regulation include unintended harm to individuals and society, difficulty in holding developers and users accountable for system harms, lack of trust in AI technologies, a lag in the AI market, misuse of AI technologies, and exacerbation of existing inequalities and societal issues.

Addressing the lack of regulation requires:

- Establishing of ethical guidelines for responsible AI development and use,
- Enforcing of data privacy regulations to protect user data and ensure ethical data usage,
- Safeguarding AI developers' intellectual property rights to foster innovation and investment,
- Clarifying liability for AI system mistakes and harm,
- Ensuring transparency and explainability by demanding clear explanations of AI system workings and decision-making processes,
- Harmonizing international regulations to avoid confusion,
- Establishing independent oversight bodies or review processes.

Although laws provide a framework for governing AI systems, they are slow to adapt to new developments and may not fully address ethical concerns. In contrast, AI ethics offers a more flexible and extensive framework for promoting responsible development and use of AI systems. It covers legal compliance and ethical considerations, such as fairness, accountability, transparency, and human values. So, AI ethics should take precedence over laws in AI development. Proactive and voluntary ethical approaches and collaboration are essential when legal bases are insufficient. PAVE , which is a coalition of more than 80 organizations committed to raising awareness and promoting the safe development and deployment of autonomous vehicles, is an excellent example for such collaboration.

Fig. 2 How to address the lack of regulation?

Information Society and AI

Information Society and Ethics of Internet

The 3rd and 4th Industrial Revolutions have led to the emergence of cyberspace as our living space. Cyberspace now involves the integration of cyberspace, individuals, and devices, creating networked society where Internet of Things (IoT), Cloud, Big Data, and Mobile technologies are seamlessly integrated.

The information society based on networked society is characterized by several key aspects:

- Shift from quantity-based to quality-based focus,
- Overcoming temporal and spatial constraints through ICT advancements,
- Ambiguous regional and national boundaries,
- Mediated forms of communication through various mediums,
- Blurring of lines between production and consumption.

ICT advancements have led to blurred lines between virtual and real worlds. This has resulted in intensified ethical concerns in the information society and the development of internet ethics. Professor Duncan Langford extensively explored the ethical implications of internet and digital technology use, emphasizing principles such as privacy, freedom of expression, and responsible use of resources. Challenges in promoting ethical behavior online include cybercrime, intellectual property issues, and online privacy concerns. The emergence of the intelligent information society, driven by AI, has come about alongside pre-existing issues of the information society that have yet to be resolved. The lack of clear leadership, absence of AI adoption strategies, and insufficient guidelines for deployment contribute to ethical dilemmas. Therefore, it is crucial to address these concerns through proactive policies and guidelines, in contrast to the reactive approach seen in internet ethics.

Digital Divide

The digital divide refers to the unequal distribution of benefits of digital technologies among social groups, geographical areas, and economic classes. It exacerbates inequalities, hinders economic and social growth, and reinforces power imbalances.

The root causes include inadequate access to technology and infrastructure, lack of digital literacy, uneven distribution of resources, language barriers, and shortage of funding and investment. Challenges in accessing technology encompass limited availability of essential hardware like computers, unreliable electricity supply, high costs, licensing restrictions, and limited funding for AI tools and resources. Insufficient training and education further impede participation in online activities, widen the knowledge gap between urban and rural areas in AI technologies, and limit access to news, information, and resources. Language barriers also pose difficulties in developing AI systems for diverse languages and dialects.

Digital divide has gender-specific implications. Women face difficulties in accessing and utilizing digital technologies due to cultural and social norms that impede their access to education and job opportunities. Biases and stereotypes further restrict women's involvement in technology fields. Focusing on specific areas such as infrastructure, healthcare, and education is crucial in bridging the digital divide. Expanding access to technology requires efforts to improve high-speed internet availability, energy infrastructure, and connectivity in underserved areas. Collaborative initiatives such as the Alliance for Affordable Internet (A4AI)[3] and Starlink[4] can contribute to reducing the cost of broadband internet and providing affordable high-speed connectivity.

[3]Alliance for Affordable Internet, "Affordability Report 2020."
[4]Starlink. "Order Starlink."

Fig. 3 Artificial Intelligence for Social Good, 2019 United Nations ESCAP & ARTNET

Some recommendations for governments to deal with digital divide:

- Install robust internet infrastructure and recruit skilled public servants.
- Provide incentives for AI technology development.
- Enhance digital literacy via education and training.

Harmful Content

Regrettably, the widespread availability of the internet has led to a rise in harmful information, encompassing content that violates laws, endangers national interests, breaches human rights, and poses harm to minors. Several characteristics of the internet, such as anonymity, direct bidirectionality, concurrency, ease of access, volatilization, replicability and modifiability of information, contribute to the widespread dissemination of harmful content. Moreover, advanced AI technologies have further facilitated the spread of harmful content, including AI-generated disinformation including deep fakes and hate speech.

To address the issue of harmful content, policymakers should strive to create a sustainable cyber ecosystem by balancing freedom of expression with preventing the dissemination of harmful content. Recommendations for policymakers include:

- Educate users about the potential negative effects of AI-generated content through awareness programs like AI4ALL[5].
- Increase access to information and knowledge to empower users in navigating and understanding the risks associated with AI-generated content.
- Utilize AI systems to combat misinformation and disinformation, such as AI tools and chips developed by companies like Facebook[6] and Google[7].
- Implement laws and regulations that hold platform operators accountable for preventing the spread of harmful content.
- Promote the development of diverse and independent media outlets creating an enabling environment for ethical reporting on AI.

Personal Information Protection

Personal information, including data that can identify individuals such as name, address, and photo, requires robust protection. The advancement of personal information protection includes the implementation of stronger encryption protocols, frequent security audits, employee education on data security and privacy, restricted access to sensitive information, comprehensive incident response plans, and the adoption of biometric and multi-factor authentication technologies.

Technological progress blurs the boundaries between life and death, which impacts data protection standards. The absence of a clear demarcation raises ethical considerations regarding the use of digital means to revive deceased individuals, including the impact on an individual's legacy, grief process, and acceptance of death. Hence, consensus and action are needed to manage the digital assets and footprints of the deceased. The GDPR in the EU provides measures such as the "Right to be forgotten" to control past data.

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[5] https://ai-4-all.org/.
[6] https://about.fb.com/news/2021/12/metas-new-ai-system-tackles-harmful-content/.
[7] https://respect.international/content-safety-api/.

AI systems have the potential to expose personal privacy information and gain extensive knowledge about individuals' lives. There are two emerging challenges: Firstly, storing personal information in robots and utilizing cloud services increase the risk of leakage, even with encryption and authentication technologies. Secondly, training AI systems with unstructured data carries the risk of sensitive information leakage. The "Ieeruda" case in Korea highlights the need for safeguards and responsible use of personal data.

Recommendations for improving data protection systems to prevent the mishandling of personal information include:

- Creating comprehensive privacy laws,
- Exercising caution when selecting and preprocessing unstructured data for AI system training,
- Providing clear explanations for personal data collection,
- Conducting regular audits and monitoring to address vulnerabilities and biases in AI systems,
- Incorporating privacy impact assessments and accountability measures that cover both structured and unstructured data.

Copyright and Infringement

Copyright is a legal right granted to individuals for expressing their original ideas and opinions. The purpose of copyright law is to protect the rights of creators of original works, including literary, musical, and artistic creations.

Determining copyright ownership and updating copyright laws to account for AI-generated works are intricate issues that require attention. AI artwork creation, particularly through Generative Adversarial Networks (GAN), raises concerns about copyright infringement during the learning process and replication of copyrighted content. Different countries are implementing laws to tackle copyright issues related to AI-generated works. The European Union assigns copyright to individuals who make creative choices, while the United States generally grants copyright to the person controlling the AI system. Encouraging research at the intersection of AI and intellectual property is important, and policymakers play a role in promoting new research and safeguarding works created using AI technologies.

Cyber Crime and Security

Cybersecurity encompasses various measures, policies, and practices aimed at safeguarding computer systems, networks, and programs from digital attacks. Cyber crimes can be classified into network intrusion, network misuse, and illegal content. Network intrusion involves hacking, service disruption, and malware. Network misuse includes internet fraud, financial cybercrime, theft of personal information, and copyright infringement. Illegal content crimes often involve the use of "deepfake" technology to create fabricated videos defaming individuals.

Developing countries encounter obstacles in the realm of cybersecurity, including weak infrastructure, limited access to secure resources, insufficient technical expertise, government surveillance and privacy, and economic inequality. Ethical concerns related to security include accountability for personal data breaches, vulnerability to cybercrime and online fraud, privacy violations through surveillance technology, and the potential misuse of biometric data. Voter registration systems and the importance of digital literacy skills further raise concerns regarding privacy, security, and potential voter suppression.

A report called "The Malicious Use of Artificial Intelligence", which was produced in February 2017 with contributions from academia, civic organizations, and industry, including Oxford University in the UK, presents unique challenges to cybersecurity due to AI's characteristics such as duality, efficiency/expandability, superiority, anonymity, affordability, and vulnerability. In contrast, security comprises three essential components: confidentiality, integrity, and availability. AI can enhance cybersecurity through various approaches, including basic security, integrated security, big data security, and AI security. Integrated security and AI work together to proactively respond to threats, analyze data, combat evolving malware, detect intrusions, identify events in real-time, and monitor global threats.[8 9 10] It is crucial to address safety risks and security risks throughout the cycle of AI systems.

To address security challenges, policymakers should consider the following recommendations:

- Adopt secure data sharing methods,
- Develop AI technologies with a focus on data security and privacy,
- Invest in cybersecurity infrastructure and expertise,
- Enhance data security and privacy knowledge and skills.

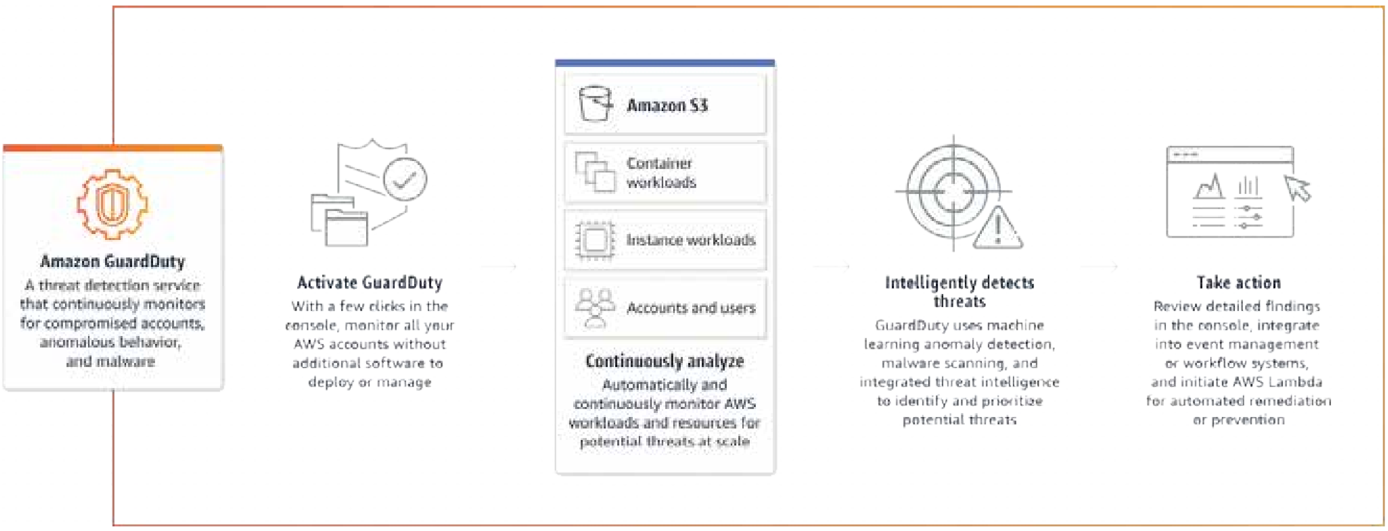

Fig. 4 Amazon GuardDuty, Example of a security monitoring service to monitor data security and privacy, Amazon Web Services

[8]https://exchange.xforce.ibmcloud.com/
[9]https://www.nortonlifelock.com/us/en/newsroom/press-releases/2018/symantec_0415_01/
[10]https://aws.amazon.com/ko/guardduty/

Intelligent Information Society and Ethics

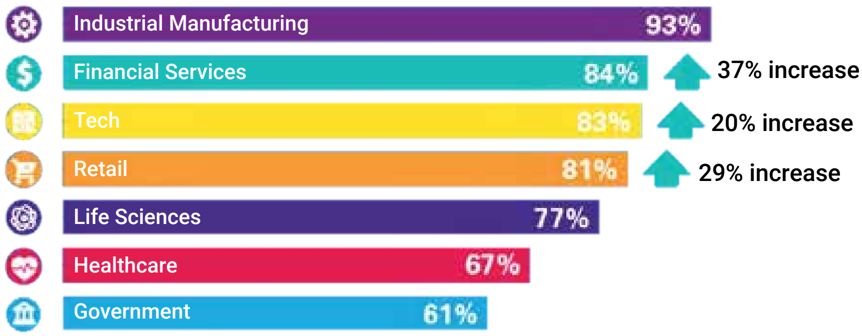

Fig. 5 Rate of AI adoption Skyrocketed during COVID-19, Business leaders and government decision-makers say AI is at least moderately to fully functional in their organization, KPMG 2021

Intelligent Information Society (IIS) and the Ethics of Data

The IIS is the convergence of humans and computers/objects for accessing and utilizing knowledge and information.

It distinguishes itself from the Information Society (IS) through:

- Advanced information technologies like AI, ML, and big data,
- Individual-centric approach with tailored intelligent systems,
- Data-driven decision making.

However, stakeholders face challenges in matching the speed of technological advancements[11]. Especially, ChatGPT has led to widespread news about AI's ability to communicate and write like humans. While ChatGPT demonstrates impressive abilities, the fact that ChatGPT is merely an extension of services used for over a decade, such as autocompletion, alleviates concerns about super intelligent entity. However, the gap between human knowledge and AI knowledge may narrow with the development of sensory sensors. AI robots like FRIDA[12] showcase advanced capabilities, but challenges still remain in areas such as creativity.

The potential of the IIS lies in enabling natural and convenient access to information, enhancing human intelligence, interconnecting products and services, removing industry boundaries, and revolutionizing people's lives. On the other hand, key concerns include the complexity of AI models and the significance of data quality.

In the IIS, Data plays a role similar to that of oil during the Industrial Revolution. A society driven by data prioritizes knowledge over traditional production factors like labor and capital. Understanding big data involves considering its volume, velocity, and variety, namely 3V's. Structured, semi-structured, and unstructured data provide different forms of information, and diverse datasets, including unbiased data from social media, and have significant business value.

During the ML process, dysfunctional problems can arise in supervised and unsupervised learning due to human involvement in data collection. Data ethics play a vital role in addressing dysfunctional problems stemming from data, defining guidelines for users to understand and adhere to data quality standards that prevent damage caused by problematic data.

Five factors contribute to data quality, namely data diversity, data accuracy, relevant data, sufficient data, and data transparency.

Government leaders play a critical role in the development of AI systems by ensuring the quality of training data to achieve accurate and unbiased outcomes:

- Promote open data and diverse datasets.
- Promote collaboration and sharing best practices,
- Balance regulation and promotion.

[11]KPMG, "AI adoption accelerated during the pandemic but many say it's moving too fast: KPMG survey 2021," 2021. [Online]. Available: https://info.kpmg.us/news-perspectives/technology-innovation/thriving-in-an-ai-world.html
[12]https://www.youtube.com/watch?v=vdMnAUtetAE

Bias and Discrimination

The challenges of fairness in AI systems stem from root causes such as insufficient diverse and comprehensive training data, cultural and social biases of developers and trainers, and the disregard for local values and norms. These factors contribute to biased outcomes, perpetuating inequalities, hindering progress, innovation, and trust in AI systems, reinforcing existing stereotypes[13].

Examples of bias include algorithmic and gender bias, biases in facial recognition and natural language processing algorithms, biased healthcare algorithms based on data from developed countries, biased outcomes in credit scoring and AI-powered recruitment systems, and biases in media and predictive policing algorithms. Policy makers should focus on minimizing biased outcomes in AI systems and establish effective remedies against discrimination. Enhancing gender equality in AI is particularly important to ensure fairness and non-discrimination. This involves protecting the rights and freedoms of girls and women throughout the AI system life cycle, creating gender-sensitive policies, eliminating gender gaps, and addressing biases and stereotypes in AI systems. To address data bias, it is crucial to standardize data collection and management systems.

Recommendations for addressing bias in local AI implementation:

- Develop representative and diverse datasets within the local context,
- Invest in a robust data infrastructure,
- Adopt standardized data formats and metadata,
- Guarantee data provider's rights,
- Create context-specific ethical guidelines,
- Conduct regular local audits and evaluations,
- Promote diversity in hiring practices.

[13]H. Devlin, "AI programs exhibit racial and gender biases, research reveals," The Guardian, 2017. [Online]. Available: https://www.theguardian.com/technology/2017/apr/13/ai-programs-exhibit-racist-and-sexist-biases-research-reveals, Accessed: Jul. 6, 2018.

Responsibility and Accountability

As AI becomes increasingly widespread, determining accountability for the actions and decisions of AI systems is of utmost importance, particularly in cases involving harm or significant consequences. The legal framework addressing the responsibility for AI system decisions must consider factors like human decision-making and public opinions, using such as the MIT Moral Machine platform that deals with potential ethical dilemmas AI may encounter. While developers and deployers of AI systems hold responsibility for their actions, granting legal personality rights and responsibilities to AI systems without accounting for human qualities raises concerns. Hence, impact assessment, due diligence mechanisms, oversight, and audit should be developed to ensure accountability and address any conflicts with human rights norms and cyber environmental well-being.

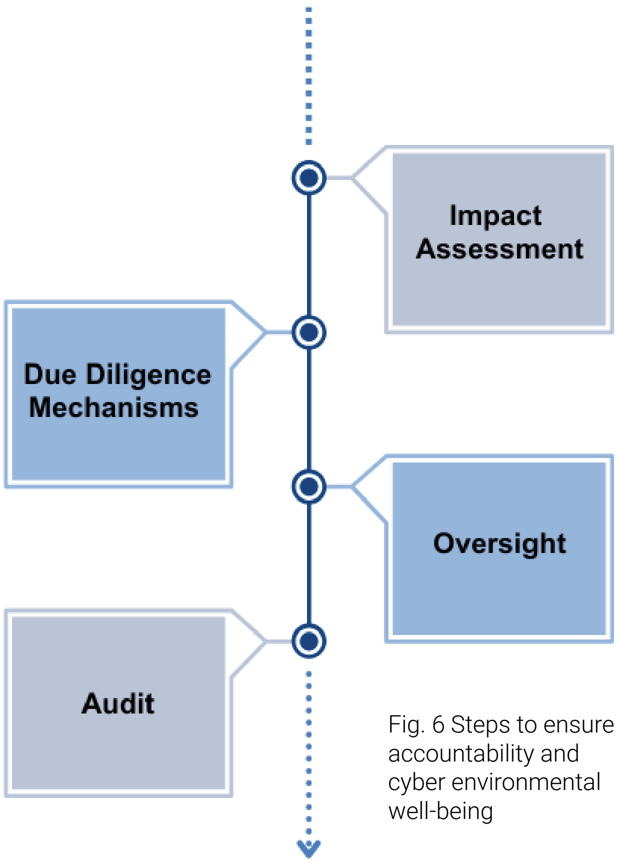

Fig. 6 Steps to ensure accountability and cyber environmental well-being

Transparency and Reliability

AI algorithms often lack transparency, making it challenging to understand their workings and identify responsible parties for errors and biases. Thus, transparency and reliability are essential for building trust in AI systems. Transparent systems with clear explanations foster trustworthiness, and reliability ensures understandable outcomes and addresses biases. Providing explanations upon request helps address human biases and decision-making processes.

Balancing personal data privacy, security, and transparency is crucial in the digital age. Maintaining a balance between these aspects is necessary to safeguard sensitive information and foster trust between individuals and organizations.

Fig. 7 transparency and reliability are essential for building trust in AI systems, 2023 United Nations ESCAP

Job Displacement

While the adoption of AI has a significant impact on job loss in developed countries, concerns also arise regarding AI replacing low-skilled jobs in developing countries. This problem can be attributed to several factors, including inadequate investment in human capital, limited opportunities for reskilling, unequal distribution of benefits from AI, and the absence or insufficiency of worker support policies.

AI automation poses a risk of widespread unemployment[14] and wage stagnation, leading to potential financial inequality and social unrest. Brain drain and the loss of skilled workers can also contribute to social and political instability. Ethical norms and values in the labor market need to be examined in response to job losses.

Issues such as the relationship between technology and labor, conflicts with new business models, and the non-face-to-face gig economy are prevalent in the labor market and raise questions regarding AI ethics. Addressing these issues is crucial to ensure that AI enhances labor productivity and creativity, while also protecting workers' rights and establishing a fair and sustainable economic system is important.

To address job displacement, key considerations include investing in education and training programs for a future-ready workforce, fostering public-private partnerships to create job opportunities and provide training, and supporting displaced workers through unemployment benefits, job search assistance, minimum wage regulations, and labor protections. Moreover, equal access to training and reskilling opportunities should be guaranteed. Efforts such as Indonesia's Pre-Employment Card program[15], India's Pradhan Mantri Kaushal Vikas Yojana (PMKVY) program[16], and expanding online programs and partnerships in the Philippines[17][18],are some good examples.

Government leaders should assess and address the impact of AI on labor markets and educational requirements:

- Encourage and support researchers to analyze the impact of AI systems on the local labor environment,
- Work collaboratively with private sector companies and civil society organizations to ensure a fair transition for at-risk employees,
- Support collaboration agreements between educational institutions and industry stakeholders to bridge the gap in required skillsets,
- Implement measures that promote fair competition in labor markets, such as establishing a minimum wage and other social safety measures.

[14]https://www.ft.com/content/7dec4483-ad34-4007-bb3a

[15]https://www.researchgate.net/publication/360078764_Kartu_Prakerja_Pre-Employment_Card_Policy_and_Its_Impact_on_Economy_and_Community_Income

[16] "Impact Evaluation of Pradhan Mantri Kaushal Vikas Yojana (PMKVY) 2.0."

[17] "ADB to Invest in Upgrading Filipinos' Skills for Better Jobs," The Asian Development Bank, Dec. 2022. [Online]. Available: https://www.adb.org/news/adb-invest-upgrading-filipinos-skills-better-jobs

[18] "ADB Study Calls for Skills Training Reform in the Philippines," The Asian Development Bank, Mar. 2021. [Online]. Available: https://www.adb.org/news/adb-study-calls-skills-training-reform-philippines

The Ethics of AI

The field of AI ethics emerged with the term coined by Nick Bostrom and Eliezer Yudkowsky from the University of Oxford. Many studies have introduced a variety of values, principles, and policy actions. These include the IEEE's "Global Initiative for Ethical Considerations in AI and Autonomous Systems" (2016), the Twenty-Three Asilomar AI Principles (2017), the European Union's "Ethics Guidelines for Trustworthy AI"(2019), and the UNESCO Recommendation on AI Ethics (2021). The UNESCO Recommendation presents ten principles encompassing proportionality, fairness, safety, sustainability, privacy, human oversight, transparency, responsibility, awareness, and multi-stakeholder governance. Policy areas cover ethical impact assessment, governance, data policy, development and cooperation, environment, gender equality, cultural heritage, education and research, communication and information, economy, labor, health, and social well-being.

The Seoul PACT provides ethical principles for AI considering mankind's prosperity, societal changes, autonomy, and intelligence. These principles include publicness, accountability, controllability, and transparency. According to the principles, ethical guidelines for developers, suppliers, users, and governments are suggested. The following is part of the guidelines for the government.

Dual Role of Government

To boost the competitive edge of AI in developing countries, vital strategies encompass making investments in research and development for utilizing large-scale AI capabilities. It is also crucial to prioritize securing data sovereignty and promoting the activation of AI application industries to be independent on foreign platforms. Moreover, they should develop AI business models using domestically specialized data to embody cultural and traditional elements to guarantee the competitiveness of domestically developed AI models, and also avert the possibility of prejudiced algorithms or data being introduced from foreign cultures.

Considering AI ethics together with AI competitiveness, government leaders and policy makers have a dual role as both developers and regulators. In their capacity as developers, they are responsible for promoting the creation and broad dissemination of technologies that serve the needs and interests of humanity. However, as regulators, they must also create explicit laws and regulations, accompanied by ethical guidelines to ensure responsible and ethical AI use. Regulating every AI industry that is rapidly evolving is impractical, so it is wise to establish a civic society that operates on principles that can evaluate and provide feedback on services that utilize AI. Then they can establish effective governance by monitoring and evaluating the deployment of AI, and conducting regular audits.

Fig. 8 The Seoul PACT